

Arbitrage Is Not a Theorem — It’s a Trade

Figlewski on Derivatives Pricing in Imperfect Markets

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1 The Seductive Lie of “No-Arbitrage”

In textbooks, *no-arbitrage* sounds like a law of nature. In reality, it is a **behavioral and institutional achievement**.

Stephen Figlewski’s core message — repeated across decades of work — is blunt:

Prices are not enforced by equations. They are enforced by trades.

“No-arbitrage” in theory is a *global consistency condition* on prices. “No-arbitrage” in practice is a **sequence of capital-constrained, risk-bearing actions** taken by real agents operating under frictions.

This distinction matters because the entire edifice of derivative pricing rests on it.

2 Arbitrage as a Control System

2.1 The Feedback Framework

Think of a derivative market as a **feedback system**:

Component	Description
Signal	Price discrepancy between derivative and replicating portfolio
Controller	Arbitrageur
Actuator	Trades in underlying + derivative
Constraints	Capital, margin, transaction costs, liquidity, timing
Noise	Volatility estimation error, jumps, discrete trading

In theory, the controller has:

- Infinite bandwidth (continuous rebalancing)
- Zero cost
- Perfect state information

In reality, every one of those assumptions fails.

2.2 The Consequence

The feedback loop is **slow, noisy, asymmetric, and capacity-limited**.

Result:

Prices hover within **arbitrage bands**, not at a point.

3 Case Taxonomy: When Arbitrage Works

Figlewski's four-case classification is one of the most useful organizing frameworks in derivatives theory.

3.1 Case 1: Arbitrage Easy in Theory *and* Practice

Example: Cash-and-carry futures on storable commodities or FX.

- Underlying is tradable
- Shorting is feasible
- Carry costs are observable
- Rebalancing is minimal

Result: Tight pricing, narrow bands.

3.2 Case 2: Arbitrage Easy in Theory, Brutal in Practice

Example: Black–Scholes delta-hedging of options.

- Requires continuous rebalancing
- Requires known volatility
- Requires frictionless trading

None of which exist. This is the core message of *Options Arbitrage in Imperfect Markets* (1989).

3.3 Case 3: Arbitrage Impossible, Law of One Price Limpes Along

Example: American vs European calls near expiration in illiquid markets.

You *can* reason about dominance, but you can't execute the textbook trade.

3.4 Case 4: Arbitrage Meaningless

Example: VIX, weather derivatives, real options, structural credit models.

No tradable underlying $\Rightarrow$ no replication $\Rightarrow$ no risk-neutral pricing.

4 The Dirty Mechanics of Option “Arbitrage”

4.1 The Black–Scholes Delta Hedge

In theory:

- Hedge eliminates risk
- Portfolio earns risk-free rate
- Price must equal model value

In practice:

- Rebalancing is discrete
- Volatility is unknown and drifting

- Transaction costs accumulate *path-dependently*
- Capital is tied up under margin constraints

4.2 Figlewski’s Monte Carlo Results (1989)

Finding	Magnitude
Daily rebalancing residual risk	~6–7% annualized
Shares traded per share hedged	~5 shares
Required bid-ask spread to break even	>30% of option value

That’s not arbitrage. That’s *manufacturing risk while pretending it vanished*.

5 Why Implied Volatility Is Not a “Forecast”

One of the most misunderstood ideas in finance:

Implied volatility is not a belief. It is a price.

If arbitrage is weak, IV reflects:

- Supply–demand imbalances
- Inventory constraints
- Hedging difficulty
- Capital scarcity

5.1 The Figlewski & Canina Evidence

In the OEX market of the 1980s:

- IV had **near-zero predictive power** for realized volatility
- Regression $R^2 \approx 0.002$

Not because markets were irrational — but because **the arbitrage enforcing the interpretation was infeasible**.

Once execution technology improved, the relationship strengthened. The model didn’t change. The *trade* did.

6 Arbitrage Bands, Not Fair Values

6.1 The Frictionless Model

$$P_{\text{market}} = P_{\text{model}}$$

6.2 The Real Market

$$P_{\text{model}} - C_{\text{arb}} \leq P_{\text{market}} \leq P_{\text{model}} + C_{\text{arb}}$$

Where C_{arb} includes:

- Expected transaction costs
- Risk premium for residual hedge risk
- Capital usage costs
- Liquidity and funding stress

This reframes “mispricing”:

A price is not wrong unless a *profitable, executable trade* exists.

7 Delta Hedging in an Imperfect World

The 1989 paper asks a precise engineering question:

What happens to the Black–Scholes arbitrage trade when you add real-world frictions?

Instead of algebra, Figlewski runs Monte Carlo simulations of delta hedging under:

- Discrete (daily) rebalancing
- Uncertain volatility
- Indivisible contracts
- Transaction costs

7.1 Discrete Hedging Creates Risk

Continuous hedging is a mathematical fantasy. With daily rebalancing, the “riskless” hedge becomes risky.

For 1-month at-the-money calls: residual hedge volatility $\approx$ **6–7% annually**.

Interpretation: Black–Scholes assumes a control system with infinite bandwidth. Real markets sample at low frequency $\rightarrow$ tracking error $\rightarrow$ noise.

7.2 Volatility Error Hurts Twice

Volatility enters:

1. The **price**
2. The **hedge ratio**

If volatility is underestimated:

- The option looks cheap
- The hedge is wrong
- Risk explodes

Deep OTM options become extremely dangerous under vol misestimation.

Asymmetry:

Overestimating volatility is safer than underestimating it.

This is a robustness result: convex payoffs + wrong curvature = tail risk.

7.3 Indivisibility Breaks Replication

If you must hedge with futures contracts or large lots:

- Delta cannot be matched exactly
- Hedge error accumulates

Replication becomes **quantized** rather than continuous. You get lattice noise instead of smooth control.

7.4 Transaction Costs Dominate Everything

Dynamic hedging requires:

- Frequent trades
- In the underlying
- In size

Result:

- Hedging costs can exceed the option premium
- Bid-ask spreads must be extremely wide to compensate

Conclusion:

Arbitrage bounds on option prices are very wide in practice.

Markets are not tightly pinned to model values; they float inside cost-and-risk bands.

8 Arbitrage as a Sociological Process

The 2017 paper steps back and reframes everything. Instead of asking “Is the model right?” Figlewski asks:

Is the arbitrage trade feasible?

8.1 The Classification

Regime	Description	Example
1	Arbitrage easy in theory <i>and</i> practice	FX forwards
2	Arbitrage easy in theory but hard in practice	Index options
3	Arbitrage impossible but Law of One Price applies	Illiquid American options

Regime	Description	Example
4	Arbitrage impossible and Law of One Price fails	VIX, weather

Key idea:

Pricing accuracy depends on how executable the hedge is.

No hedge $\rightarrow$ no enforcement $\rightarrow$ model drifts away from price.

9 Historical Evidence: Index Options

For S&P index options:

- Replicating portfolio = hundreds of stocks
- Continuous trading = impossible
- Costs = huge
- Liquidity = finite

Result: Implied volatility did **not** forecast realized volatility in the 1980s.

Regression: Realized $\sigma \approx$ unrelated to implied σ

Because:

Nobody could practically do the arbitrage trade.

So implied volatility was not “the market’s forecast.” It was a supply–demand artifact.

10 Systems View

Model the situation as a feedback system:

Element	Value
State	True price
Signal	Model value
Controller	Arbitrageur
Actuator	Hedge trades
Friction	Costs, risk, capital constraints

If controller gain is low $\rightarrow$ system does not track model.

Black–Scholes assumes:

- Gain $\rightarrow \infty$
- Delay $\rightarrow 0$

- Noise $\rightarrow 0$

Reality:

- Gain limited
- Delay discrete
- Noise large

So price $\neq$ model, but oscillates around it with bias.

11 Connection to Agent-Based Thinking

Arbitrage is a **locally constructive sequential game**:

- Agents act on local information
- Actions change future states
- Capital constraints bind dynamically
- Equilibrium is emergent, not imposed

This is exactly the definition of real economic systems in agent-based computational economics.

Black–Scholes is a *global fixed point*. Markets are *path-dependent adaptive systems*.

12 The Deep Unification

These papers unify three domains:

Domain	Concept
Finance	Replication
Control Theory	Tracking error
Economics	Limits to arbitrage

Together they imply:

A derivative is not priced by mathematics alone. It is priced by the weakest link in its replication chain.

If replication is:

- **Easy** $\rightarrow$ model dominates
 - **Hard** $\rightarrow$ order flow dominates
 - **Impossible** $\rightarrow$ model collapses
-

13 Why This Matters for Modern Modeling

This perspective explains:

- Why stochastic volatility and jumps matter
- Why replication $\neq$ valuation
- Why American options behave “wrongly” near expiration
- Why power markets, weather, congestion, and real options **cannot** be priced by naive arbitrage logic

And it suggests a better architecture:

Prices emerge from constrained agents optimizing under uncertainty — not from axioms.

That is the bridge between:

- Derivatives theory
 - Agent-based models
 - Data assimilation
 - Real-world trading systems
-

14 Final Synthesis

Figlewski’s message across 30 years:

Black–Scholes is not wrong. But it is incomplete.

It prices under:

- Infinite liquidity
- Infinite speed
- Infinite capital
- Perfect information

Markets price under:

- Discrete time
- Uncertain volatility
- Margin constraints
- Costly trades
- Fragile liquidity

Therefore:

Derivatives are priced by the struggle between geometry (payoff structure) and plumbing (trading reality).

The model draws the map. The arbitrageur drives the road. And the road has potholes.

15 Summary

15.1 One Sentence to Keep

Arbitrage is not a proof technique — it is a fragile, capital-intensive control process embedded in time.

Once you see that, a lot of “anomalies” stop being anomalies and start being *diagnostics*.

15.2 Key Insights

Insight	Implication
Arbitrage is an activity, not a theorem	Prices float within bands
Delta hedging creates residual risk	“Riskless” portfolios are risky
Implied volatility is a price, not a forecast	IV reflects supply/demand, not beliefs
Execution technology matters	Better trades → tighter pricing
No hedge means no enforcement	Model $\neq$ market without arbitrageurs